University of Bologna
Second cycle degree programme (LM) in Quantitative Finance
Course on Advanced Interest Rate Models and Markets
Marco Bianchetti
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1. Problems

Solve the problems assigned below.

2. Market data

The market data to complete the problems are available in the associated file MarketData31Oct2019.xls. Each market data is described in the course slides and was discussed during the course. The market data refer to a specific date, 31 October 2019, which is also the valuation date to be used for the exercises.

3. Approach to problems and solutions

The problems have been designed according to real market conditions, i.e. they refer to real financial instruments, market data, and models. Thus, each problem should be approached considering the relevant market context, including conventions and any other detail. Possible under-specifications in the problems could require some additional assumption to obtain a well-defined solution. Finding the reasonable assumptions necessary to solve the problem is part of the problem itself (as in the real life).

4. Deliverables

Carefully check the deliverables indicated for each problem and report exactly what is required. Explain carefully what you have done: main steps, reference to spreadsheet and or code, results, comments and considerations.

5. Teams' interaction

During the examination, members of the same team are encouraged to work together, sharing and cross-checking questions, solutions, spreadsheets and codes. Interaction is allowed only among members of the same team. Cross-team interaction is forbidden and will be considered as cheating, penalizing the whole group.

6. Discussion

Each Group will be asked to present the results in a live session. During the session, each member of the Group will be examined on any deliverable assigned to the Group (i.e. all the members are required to know anything in the delivered homework) and on the theory behind the exercises reported in the course material.

7. Valuation

The valuation will be based on the following criteria:

- number of problems solved;
- correctness of solutions and of numerical results;
- clarity of exposition, usage of the technical terminology;
- spreadsheets and codes: order, clarity, comments;
- codes: correct usage of the programming language, order, clarity, comments;
- oral exposition: knowledge of the homework and of the corresponding theory.

Wrong, non-working, confused, non-commented deliverables will be penalized. Clearly copied problems will be strongly penalized.

PROBLEMS			
#	**Problem**	**Instruments**	**Deliverable**
1.	**Swaption cubes**: a) from the Swaption Physical Settled forward premium cube quoted on the EUR market, compute the corresponding normal (Bachelier) and shifted-lognormal (Black) implied forward volatility cubes, and compare them for different values of the lognormal shift. Discuss possible cases where the numerical inversion of the Bachelier or Black formulas is problematic. b) Using the volatility cubes above, compute the corresponding Swaption Cash IRR forward premium cube.	Spreadsheet, code, market data sheets.	Spreadsheet + code with relevant data, calculations and charts, written relation with explanations and comments.
2.	**SABR**: a) implement the shifted SABR model (Hagan et al. “Universal Smiles”, 2016) for Swaptions and calibrate its parameters to the market physical Swaption volatility cube;	Spreadsheet, code, market data sheets.	Spreadsheet/code with relevant data, calculations and charts, written relation with explanations and comments.

	b) test the calibration precision with respect to different objective functions (i.e. relative price differences, RMSE volatility, vega-weighted RMSE volatility) c) test different optimization methods in terms of calibration precision and performance d) test SABR parameter redundancy.		
3.	**Artificial Intelligence**: use a public Generative Artificial Intelligence to solve the previous exercises and produce the same deliverables above. The exercise is not about the perfect reproduction of the work above, but rather to show the extent to which artificial intelligence can help or hinder the work.	Public Generative Artificial Intelligence (e.g., ChatGPT and/or others)	GenAI inputs and outputs, analyses and comments. The idea of the exercise is not perfect reproduction but to show how much AI may help or not the task.